

# todo Co-Formation Games

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**Abstract—**todo

**Index Terms—**Collective dynamics, Game theory, Network formation games, Network theory (graphs)

$$f_i^c(a) = -\alpha_c \|a_i\| - \frac{1}{\|k\|} \sum_{j \in K_i} |x_i - x_j| \quad (3)$$

## I. INTRODUCTION

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### A. Paper overview

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## II. METHODS

### A. Definitions & Equations

TABLE I  
LIST OF VARIABLES USED IN THIS DOCUMENT

| Variable                   | Symbol        | Domain                  |
|----------------------------|---------------|-------------------------|
| Number of nodes            | $n$           | $\mathbb{N}$            |
| Node of concern            | $i$           | $\{1, \dots, n\}$       |
| Set of $i$ 's neighbours   | $K_i$         | $\mathbb{N}^{ k_i }$    |
| Network graph <sup>1</sup> | $G$           | $\{0, 1\}^{n \times n}$ |
| Opinion                    | $x_i$         | $\mathbb{R}$            |
| Action profile             | $a_i$         | $\{0, 1\}^n$            |
| Costs per edge             | $\alpha_c$    | $[0, \infty[$           |
| Pairwise distance          | $d_{ij}$      | $\mathbb{N}$            |
| Network valuation          | $v$           | $] -\infty, 0]$         |
| Network expenses           | $f^c$         | $] -\infty, 0[$         |
| Cost function              | $c$           | $] -\infty, 0[$         |
| Communication rate         | $r_c$         | $[0, \infty[$           |
| Formation rate             | $r_f$         | $\{0, 1\}$              |
| Convergence threshold      | $\varepsilon$ | $[0, \alpha_c]$         |

where a norm  $\|\cdot\|$  is the L0 norm, returning the count of non-zero elements.

#### a) Equations:

- Cost function

$$c_i = v_i(a) + f^c(a) \quad (1)$$

- Valuation function

$$v_i(a) = - \sum_{j=1, \dots, n} d_{ij} \quad (2)$$

- Network expenses:

#### b) Algorithm:

```

1 while ( $t < t_{\max} \wedge \varepsilon < \Delta c$ )
2   if  $\text{mod}(t, r_c) = 0$  then
3     select random node  $i$ 
4      $x_i$  arrow.l get opinion update  $x_i^u$ 
5   end if
6   if  $r_f = 1$  then
7     select random node  $i$ 
8      $a_i$  arrow.l get action update  $a_i^u$ 
9   end if
10  get new costs  $c_{t+1}$ 
11   $\Delta c = |c_{t+1} - c_t|$ 
12   $c_t \leftarrow c_{t+1}$ 
13   $t \leftarrow t + 1$ 
14 end while

```

Algorithm 1. Dynamics of our synchronous move game

## III. OLD STUFF FROM THE TEMPLATE



Fig. 1. A circle representing the Sun.

In Fig. 1 you can see a common representation of the Sun, which is a star that is located at the center of the solar system.

In Table II, you see the planets of the solar system and their average distance from the Sun. The distances were calculated with (1) that we presented in Section II.

<sup>1</sup>The graph is represented as an adjacency matrix.

TABLE II  
THE PLANETS OF THE SOLAR SYSTEM AND THEIR AVERAGE DISTANCE FROM  
THE SUN

| Planet  | Distance (million km) |
|---------|-----------------------|
| Mercury | 57.9                  |
| Venus   | 108.2                 |
| Earth   | 149.6                 |
| Mars    | 227.9                 |
| Jupiter | 778.6                 |
| Saturn  | 1,433.5               |
| Uranus  | 2,872.5               |
| Neptune | 4,495.1               |

REFERENCES